

Ontogeny of the Adrenal Gland in the Spiny Mouse, With Particular Reference to Production of the Steroids Cortisol and Dehydroepiandrosterone

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Synthesis of the androgen dehydroepiandrosterone (DHEA) by the fetal adrenal gland is important for placental estrogen production and may also be important for modulating the effects of glucocorticoids on the developing brain. The presence of cortisol in spiny mouse (*Acomys cahirinus*) blood led us to determine whether the adrenal gland of this precocial rodent also synthesized DHEA. Cytochrome P450 enzyme 17 α -hydroxylase/17,20-lyase (P450c17), cytochrome-b5 (Cytb5), and 3 β -hydroxysteroid dehydrogenase (3 β HSD) were detected in the adrenal gland from 30 days gestation (term = 39 days), and DHEA, cortisol, and aldosterone were detected in fetal plasma from this time. Plasma DHEA concentrations increased 4-fold, whereas cortisol concentrations decreased from day 30 of gestation until the day of birth. Explant culture of fetal adrenal tissue showed that DHEA was produced from exogenous pregnenolone, and thus, the DHEA in the fetal circulation is likely to be of fetal origin. Clear zonation of the fetal adrenal cortex was evident by 38 days gestation when expression of Cytb5 was present throughout the cortex, and coexpression of P450c17 and Cytb5 occurred in the zona reticularis and fasciculata. 3 β HSD was expressed in the cortex from at least 30 days gestation and decreased as term approached, consistent with the fall of cortisol in late gestation in this species. These results show that the spiny mouse adrenal gland, like that of the human fetus, can synthesize and secrete DHEA from at least 30 days (relative gestation length, 30 days of a 39-day gestation, 0.76) of gestation, and DHEA may have important roles in placental biosynthesis of estrogens and in modulating the actions of glucocorticoids in the developing brain in this species. (**Endocrinology** 154: 1190–1201, 2013)

In addition to a glucocorticoid, in some species, the adrenal gland also produces the C₁₉ steroid dehydroepiandrosterone (DHEA) (1). DHEA is an androgen and estrogen receptor agonist (2–4) and also a selective glucocorticoid receptor antagonist (5, 6). In the placenta, DHEA sulfate (DHEA-S) is converted to bioactive estrogens that promote labor and delivery in a number of species (7, 8). In the brain, DHEA has been established as a neuroactive neurosteroid where it has trophic effects on neuronal growth and differentiation, as shown by its ability to induce morphological changes in primary cultures of

embryonic neocortical neurons, eg, by inducing increased axonal length and formation of varicosities and basket-like formations around other cell bodies (9). Perhaps significantly for the developing brain, DHEA also protects neurons against glucocorticoid-induced neurotoxicity (10–13) and other perturbations that result in oxidative stress (14). DHEA can therefore be regarded as a stress hormone and a neurosteroid that modulates the effects of glucocorticoids on the brain (13, 15, 16). Clinically, low levels of plasma DHEA relative to cortisol in neonates after maternal stress during pregnancy have been corre-

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Abbreviations: Cytb5, cytochrome-b5; DHEA, dehydroepiandrosterone; DHEA-S, DHEA sulfate; HRP, horseradish peroxidase; 3 β HSD, 3 β -hydroxysteroid dehydrogenase; P450c17, cytochrome P450 enzyme 17 α -hydroxylase/17,20-lyase; PREG, pregnenolone; TOH, tyrosine hydroxylase.

lated with the development of a number of behavioral pathologies in adulthood including depression, bipolar disorder, and schizophrenia (17–20), further suggesting the importance of DHEA on brain development of the fetus and the neonate.

Most studies demonstrating the neuroprotective and antiglucocorticoid effects of administered DHEA on the nervous system have been carried out in the rat, a species whose adrenal gland synthesizes neither cortisol nor DHEA in physiologically significant concentrations during gestation, although the rat does have the capacity to transiently secrete androgens arising from 17-hydroxylation during prepubescence (21). Based on previous findings by us (22) and others (23) showing that the spiny mouse (*Acomys cahirinus*) adrenal gland secretes the C₂₁ steroid cortisol, we asked whether this species might also synthesize and secrete C₁₉ steroids such as DHEA and, if so, when this occurs during pre- and postnatal development and what relationship it has to the development and maturation of the adrenal cortex. The *de novo* synthesis of DHEA from cholesterol depends on the presence and activity of the microsomal enzyme cytochrome P450 enzyme 17 α -hydroxylase/17,20-lyase (P450c17), and the accessory hemoprotein cytochrome-b5 (Cytb5) (24). Importantly, P450c17 and Cytb5 need to be colocalized, because the function of Cytb5 is to selectively enhance the 17,20-lyase activity of P450c17 (25, 26), specifically supporting androgen synthesis in the adrenal gland (27). Therefore, an aim of this study was to determine when these key enzymes are expressed in the developing adrenal cortex. Because the zona glomerulosa of the cortex is also the major source of the sodium-retaining hormone aldosterone, a C₂₁ steroid requiring synthesis by the enzyme 3 β -hydroxysteroid dehydrogenase (3 β HSD), we also investigated the pre- and postnatal changes in adrenal expression of this enzyme. Finally, because little is known about the development of the adrenal gland in the spiny mouse, and it is likely that cortical differentiation occurs *pari passu* with the adrenal medulla, we have also described the development of the definitive medulla and its innervation for the first time in this species.

Materials and Methods

Animals

All experimental procedures were approved by Monash University School of Biomedical Sciences Animal Ethics Committee and were conducted in accordance with the Australian Code of Practice for the Care and Use of Animals for Scientific Purposes. Spiny mice were bred and housed as previously described (28).

Blood and tissue collection

All blood and tissue collections occurred between 10:30 and 11:30 AM. Pregnant dams were killed by cervical dislocation while under isoflurane inhalation sedation (Abbott, Botany, Australia) when fetuses were 30, 35, and 38 days of gestation (average gestation length is 39 days), and maternal blood was collected by cardiac puncture. Upon excising the uterus, the pups were immediately killed by decapitation and blood was collected from trunk vessels. Due to small volumes of blood collected from each fetus, blood from all pups in each litter (2–4) was pooled. Animals aged from the day of birth until 20 days postnatal age were killed by decapitation, and blood was collected from the trunk vessels. Maternal (n = 4 per age of gestation) and fetal/neonatal (n = 9–11 per age group) plasma was obtained by centrifugation (5 minutes at 3000g at 4°C) and stored at –20°C before DHEA, cortisol, and aldosterone RIA. Plasma from fetal animals (30 days of gestation, n = 7 in total) was used for corticosterone RIA. Plasma from 8- to 20-day-old animals (n = 13 in total) was used for ultraperformance liquid chromatography tandem mass spectrometry.

Both pre- and postnatal adrenal glands were collected through an incision in the midline of the spiny mouse abdomen. The left adrenal was excised, immersed in 4% paraformaldehyde for 24 hours and processed for histology and immunohistochemistry (n = 6 per age group). Right adrenals were weighed and frozen at –80°C for immunoblot analysis (n = 4 per age group). In a separate cohort, left and right adrenals were rapidly removed from fetuses of 35 days gestation (n = 11 animals) and on the day of birth (n = 6 animals) for explant culture and steroid synthesis experiments.

Immunohistochemistry

Single label

Immunohistochemistry was carried out on 5- μ m paraffin-embedded sections. Sections were deparaffinized in xylene, and antigen retrieval carried out in 0.01M citric acid buffer (pH 6) using a microwave oven (3 periods of 5 minutes duration). The sections were then rinsed in 0.1M PBS (pH 7.4), incubated with 3% hydrogen peroxide for 20 minutes at room temperature to block endogenous peroxidase activity and then incubated with 5% normal goat serum in 0.1M PBS containing 3% BSA for 45 minutes at room temperature to block nonspecific binding. Sections were then incubated for 12 hours at 4°C with primary antibody in 0.1M PBS (Table 1). The sections were then washed 3 times in PBS and incubated for 1 hour at room temperature with the appropriate IgG secondary antibody (biotinylated anti-rabbit [P450c17 and Cytb5], anti-goat [3 β HSD], or anti-mouse [tyrosine hydroxylase (TOH), and nonphosphorylated neurofilament-H [SMI-32]) at 1:250 dilution (all from Vector Laboratories, Burlingame, California). Antibody binding was visualized using streptavidin horseradish peroxidase (HRP) (1:200; Amersham Biosciences, Buckinghamshire, United Kingdom), with metal-enhanced diaminobenzidine (Pierce Biotechnology Inc, Rockford, Illinois) as the chromogen. Sections stained for P450c17, Cytb5, and TOH were lightly counterstained with hematoxylin for 10 seconds.

Table 1. Primary Antibodies Used for Immunohistochemical Analysis

Primary Antibody	Antigen	Source	Dilution
P450c17	Polyclonal rabbit antiovine P450c17	A.J.C.	1:1000
Cytochrome b5	Polyclonal rabbit antihuman cytochrome b5	A.J.C.	1:1000
3 β HSD	Polyclonal goat antihuman 3 β HSD	Santa Cruz Biotechnology	1:100
TOH	Monoclonal mouse antirat TOH	Boehringer, Mannheim	1:1000
Neurofilament-H (SMI-32)	Monoclonal mouse antihuman SMI-32	Covance, Princeton, New Jersey	1:400
Synaptophysin	Monoclonal mouse antiovine synaptophysin	Chemicon, Billerica, Massachusetts	1:700

Double label

Sections were incubated with polyclonal antibodies to P450c17 (1:1000; A.J.C.) and Cytb5 (1:1000; A.J.C.) using a sequential immunofluorescence protocol. Sections first underwent a blocking step and were incubated with the first primary antibody followed by incubation with the first secondary antibody (Alexa Fluor 594 goat antirabbit, 1:1000; Molecular Probes Inc, Eugene, Oregon). After several washes in PBS, the sections underwent a second blocking step followed by incubation with the second primary antibody and the second secondary antibody (Alexa Fluor 488 goat antirabbit, 1:1000; Molecular Probes). Sections were then viewed using an Olympus BX41 fluorescence microscope.

Immunoblotting

Total protein was extracted from snap-frozen adrenals and concentration determined using a Pierce BCA (bicinchoninic acid) protein assay kit (Thermo Fisher Scientific, Scoresby, Australia). Protein (35 μ g) was subjected to a reducing SDS-PAGE and transferred onto a polyvinylidene difluoride membrane (Immobilon-P; Millipore, Billerica, Massachusetts). Membranes were blocked with 5% skim milk in PBS/0.5% Tween 20 before incubation with the following primary antibodies overnight at 4°C: 1:10 000 rabbit polyclonal anti-P450c17, 1:10 000 rabbit polyclonal anti-Cytb5, 1:2000 polyclonal goat anti-3 β HSD, and 1:2000 mouse monoclonal anti- β -actin (Santa Cruz Biotechnology, Santa Cruz, California). Membranes were incubated with secondary antibodies (1:5000 goat antirabbit HRP [Santa Cruz Biotechnology], 1:5000 rabbit antigoat HRP [Serotec, Oxford, United Kingdom], and 1:10 000 goat antimouse HRP [Serotec]) at room temperature for 1 hour, and the immune-labeled protein was visualized using an Immobilon Western chemiluminescent HRP substrate (Millipore).

Hormone assays

Cortisol

After extraction of plasma using dichloromethane, total plasma cortisol was measured by an RIA described previously (29), using cortisol (H-4001; Sigma Chemical Co, St Louis, Missouri) as standard. The values obtained by RIA were corrected for the extraction recovery (mean recovery, 79.9%). The inter-assay coefficient of variation was 7.2%, the intra-assay coefficient of variation was 3.4% at 1.5 ± 0.26 ng/mL and 8.39% at 64.5 ± 2.09 ng/mL. The assay sensitivity was 0.38 ± 0.1 ng/mL.

Corticosterone

A standard corticosterone immunoassay kit (Diagnostic Systems Laboratories, Inc, Gladesville, New South Wales), was used to assess plasma corticosterone. The intra-assay coefficient of

variation was 2.5%–3.9%, and the assay sensitivity was 1.6 ng/mL.

DHEA

Levels of DHEA were measured in spiny mouse plasma and explant culture media using a commercially available RIA kit according to manufacturer's instructions (DSL8900; Beckman Coulter, Brea, California). The cross-reactivity of the antisera used in the assay was determined by the manufacturer and was less than 0.75% with other structurally similar steroids. For plasma, the intra-assay coefficient of variation was 3.9% at 2.5 ± 0.75 ng/mL and 6.2% at 7.5 ± 2.25 ng/mL. The assay sensitivity was 0.009 ng/mL. For media, the intra-assay coefficient of variation was 1.67% at 1.7 ± 0.11 ng/mL and 5.7% at 10.01 ± 1.59 ng/mL. The assay sensitivity was 0.02 ng/mL. The presence of authentic DHEA was determined by ultraperformance liquid chromatography tandem mass spectrometry; details of the methods and results are given in supplemental data (published on The Endocrine Society's Journals Online web site at <http://endo.endojournals.org>).

Aldosterone

A standard RIA kit was used to measure the plasma concentration of aldosterone (Beckman Coulter). The intra-assay coefficient of variation was <4.5%. The assay sensitivity was 6.64 pg/mL.

Explant culture

Fetal ($n = 11$, 35 days gestation) and neonatal ($n = 6$, day of birth) adrenal glands were dissected free of adipose and connective tissue in sterile saline. Each adrenal was halved longitudinally and placed in a single well of a 24-well plate and cultured for 24 hours in 500 μ l of Ham's DMEM-F12 (with 10% [vol/vol] fetal bovine serum, 5% [vol/vol] horse serum, and 1% [vol/vol] penicillin streptomycin; Invitrogen, Mulgrave, Australia) at 37°C and equilibrated with 5% CO₂ in air. Twenty-four hours later, the media were changed and adrenal halves were randomly divided into two groups: media alone (control group) or media enriched with 2.5mM pregnenolone (PREG) (Santa Cruz Biotechnology) for another 24, 48, or 72 hours. At the end of the culture period, media were collected and stored at -20°C until analysis of DHEA using RIA (as above).

Data analysis and statistics

All data are expressed as mean $\pm$ SEM, and statistical significance was determined using statistical analysis software (Prism; GraphPad Software Inc, San Diego, California). Adrenal weight to body weight ratios were transformed using the log₁₀ to equal-

ize variances, and transformed data were analyzed using one-way independent-measures ANOVA, with the Tukey post hoc test. Age-dependent changes in the plasma concentrations of aldosterone, DHEA, and cortisol were analyzed using the one-way independent-measures ANOVA with the Tukey post hoc test. The concentrations of plasma cortisol relative to corticosterone at 30 days of gestation were analyzed using a paired *t* test. Adrenal explant cultures were analyzed using a two-way repeated-measures ANOVA, with the Bonferroni post hoc test for multiple comparisons. Statistical significance was accorded when $P < .05$. There were no significant differences between sexes for any parameter included in the analysis, and thus, the sexes have been combined for all of the data shown.

Results

Growth of the adrenal gland

As expected, total adrenal weight increased in absolute terms from 30 days gestation to 8 days postnatal age (Figure 1A). From late gestation and until 8 days of postnatal age, the weight of the adrenal gland relative to body weight was relatively constant (~ 0.2 mg/g), but it was significantly greater (0.57 ± 0.16 mg/g) at 35 days gestation than at all other times (Figure 1B).

At 30 days gestation, it was not possible to clearly distinguish the presumptive cortex and medulla using hema-

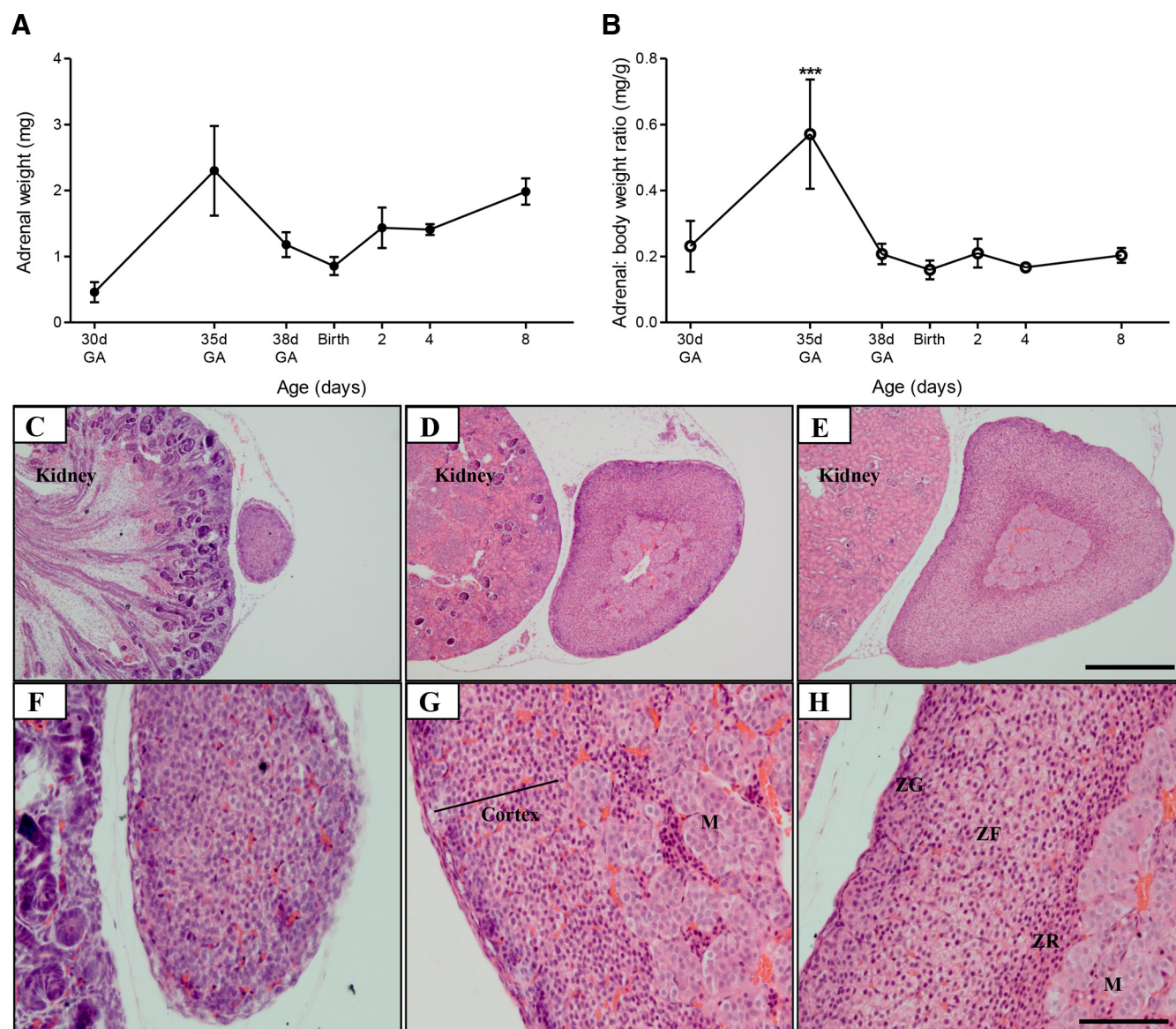


Figure 1. A and B, Total adrenal weight (A) and the adrenal to body weight ratio (B) from 30 days gestational age (GA) to 8 days postnatal age. The relative weight of the adrenal glands was significantly greater at 35 days GA compared with all other time points ($n = 9-11$ per age group; *** $P \leq .0001$). C–H, Hematoxylin and eosin staining in the male adrenal gland at 30 days gestation (C), 38 days of gestation (D), and 8 days of postnatal age (E). Note the absence of distinct cortical layers at 30 days of gestation and the presence of the zona reticularis (ZR), zona fasciculata (ZF), and zona glomerulosa (ZG) at 8 days of age. F–H, High-magnification ($\times 400$) images of panels C–E, respectively. Scale bars, 500 μm (C–E) and 100 μm (F–H). M, medulla.

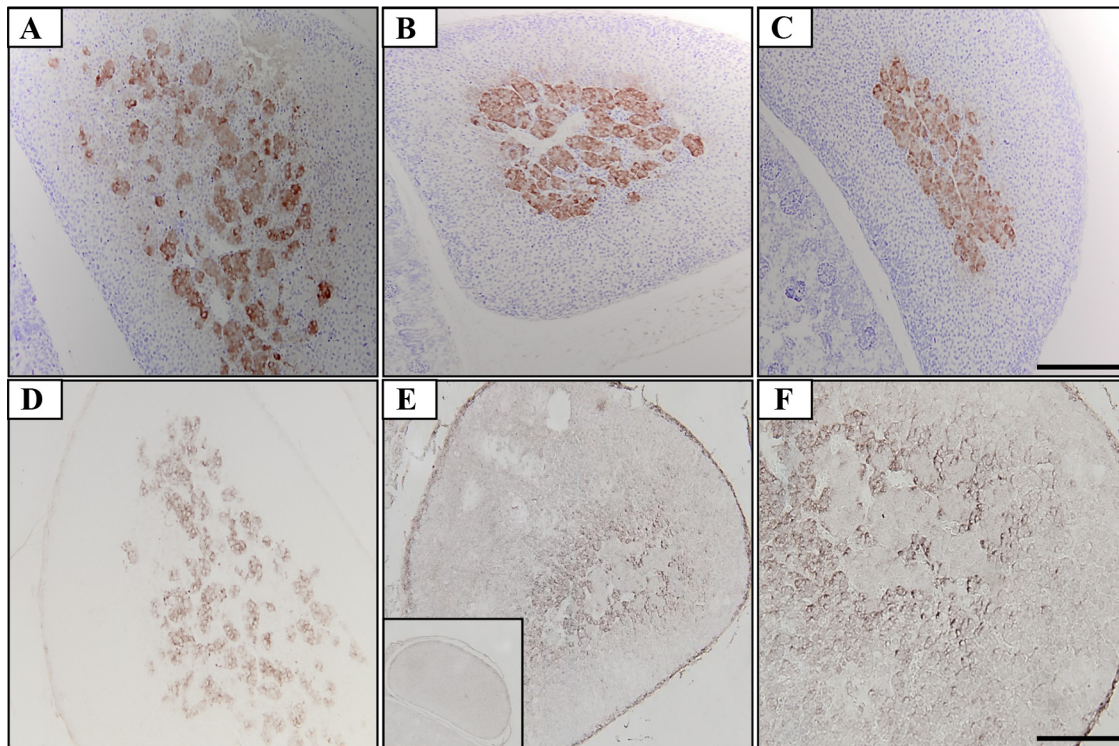


Figure 2. Differentiation of the adrenal cortex and medulla in the male fetal and neonatal spiny mouse. A–C, Adrenal chromaffin cells were identified by TOH immunocytochemistry at 30, 35, and 38 days of gestation, respectively. D, The presence of innervations at 30 days of gestation is indicated by the presence of the presynaptic protein synaptophysin. E and F, Neurofilament-H (SMI-32) immunocytochemistry shows the presence of neurofilament in the center of the adrenal gland at 30 days of gestation (inset in E shows control staining of fetal adrenal gland with no primary antibody). F, High-magnification ($\times 400$) image of E, showing neurofilament-positive cells closely surrounding the large inner TOH-positive cells. Scale bars, 200 μm (A–E) and 100 μm (F).

toxylin and eosin staining (Figure 1C), but by 38 days, these primary zones were clearly evident (Figure 1D). At 30 days gestation, vascular spaces (presumably venous sinusoids) were present throughout the gland, whereas by day 38, the largest vascular spaces were present within the medulla and at the border of the medulla and cortex (Figure 1, F and G). Using immunocytochemical localization of TOH, adrenal chromaffin cells could be identified from at least 30 days gestation (Figure 2A). TOH-positive cells were loosely distributed in the center of the gland until approximately 35 days gestation (Figure 2B) but by 38 days, they were highly compacted and there was then obvious and distinct differentiation of cortex and medulla (Figure 2C). Immunoreactivity for synaptophysin (Figure 2D) and SMI-32 (Figure 2, E and F) showed that expression of a synaptic and neurofilament protein, respectively, was present in the medulla from day 30 of gestation. The SMI-32-positive cells were localized in the center of the fetal adrenal gland, completely surrounding the clusters of the innermost, large TOH-positive chromaffin cells (Figure 2, E and F).

Differentiation of cortical zones

Zonation of the adrenal cortex was evident by 38 days of gestation, when an outer and inner rim of eosinophilic

cells could be identified (Figure 1, D and G). At 35 and 38 days of fetal age, P450c17 immunoreactivity was also identified in two inner zones, presumptively identified as the zona reticularis and zona fasciculata (Figure 3, B and C). Expression of Cytb5 occurred throughout the cortex and also in the medulla of the fetal and postnatal adrenal gland (Figure 3, H–M). P450c17 and Cytb5 were clearly expressed in the same cell in the central region of the fetal adrenal cortex from 30 days gestation, as shown by double-label immunocytochemistry (Figure 4). At 8 days of postnatal age, immunopositive cells for both enzymes were localized throughout the zona fasciculata and in the region of the zona reticularis abutting the zona fasciculata (Figure 4I).

Protein expression of $3\beta\text{HSD}$ was identified in the fetal adrenal cortex from 30 days gestation (Figure 3O). $3\beta\text{HSD}$ expression increased more greatly in the zona fasciculata relative to the zona reticularis and zona glomerulosa as term approached (Figure 3, P–R). This pattern of expression persisted into early postnatal life (Figure 3, S and T). The expression of P450c17, Cytb5, and $3\beta\text{HSD}$ in the fetal adrenal gland (cortex and medulla) was verified by immunoblot analysis. All enzymes showed immunoreactivity at expected molecular weights from 30 days of gestation (Figure 3V).

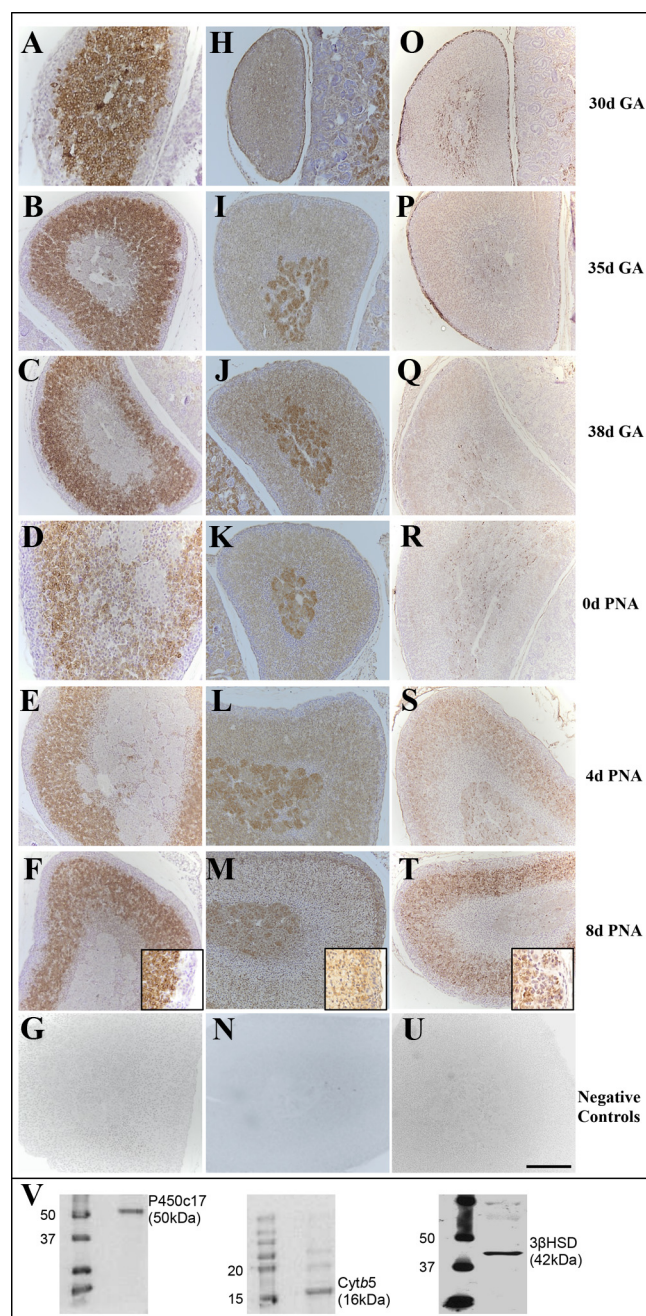


Figure 3. A–U, Cortical differentiation in the male fetal and neonatal spiny mouse adrenal gland. P450c17 expression in a 30-, 35-, and 38-day fetal and 0-, 2-, 4-, and 8-day-old postnatal spiny mouse (A–F, respectively) and of Cytochrome b5 (H–M, respectively) and of 3βHSD (O–T, respectively). Insets in F, M, and T show high-magnification ($\times 400$) images of P450c17, Cytochrome b5, and 3βHSD, respectively, in the zona glomerulosa of the 8-d-old adrenal. G, N, and U represent 4-d-old postnatal adrenal glands with no primary antibody for P450c17, Cytochrome b5, and 3βHSD, respectively. Scale bar, 200 μ m. V, Immunoblot verification of enzyme expression and antibody specificity for P450c17, Cytochrome b5, and 3βHSD in the spiny mouse fetal adrenal gland at 30 days gestational age. Abbreviations: GA, gestational age; PNA, postnatal age.

Plasma steroid levels

Maternal plasma DHEA concentrations measured from 30 days gestation (1.5 ± 0.65 ng/mL) until term

(1.4 ± 0.54 ng/mL) did not change significantly over this time (Figure 5A). The concentrations of DHEA in maternal plasma were consistently higher than fetal plasma DHEA (average fetal DHEA concentration, 0.65 ± 0.11 ng/mL). Fetal plasma DHEA concentrations increased approximately 2-fold at birth (Figure 5A), and were ~ 2 ng/mL over the first 8 days of postnatal life. Plasma cortisol concentrations were significantly higher in the fetus at 30 days gestation (309.5 ± 59.8 ng/mL) compared with 35 days (194.2 ± 10.1 ng/mL), 38 days (57.2 ± 5.20 ng/mL), and on the day of birth (70.2 ± 5.0 ng/mL), so that the fetal plasma DHEA to cortisol ratio increased from 30 to 38 days gestation, due to the decrease in the concentration of cortisol (Figure 5, B and C). The DHEA detected by RIA in spiny mouse plasma was authenticated by positive ionization mass spectrometry (see supplemental data). Plasma aldosterone concentrations increased from day 30 of gestation (9.4 ± 1.6 pg/mL) until the day of birth (183.61 ± 58.4). The levels of aldosterone decreased significantly in the early neonatal period (39.82 ± 8.2 and 21.67 ± 4.6 pg/mL at 2 and 4 days of postnatal age, respectively) before increasing and reaching 217.31 ± 78.41 pg/mL by postnatal day 8 (Figure 5C).

Biosynthesis of DHEA in the fetal adrenal gland

The bioactivity of the P450c17 enzyme in the fetal spiny mouse adrenal gland was evaluated by the ability of explants of fetal (35 days gestation) and neonatal (day of birth) adrenal tissue to convert PREG into DHEA. Culture of explant fetal adrenal glands in the presence of 2.5mM PREG resulted in a maximum synthesis and secretion of 6.43 ± 0.46 ng DHEA/ng tissue after 48 hours in culture (Figure 6). DHEA synthesis by neonatal adrenal explants reached a maximum of 9.67 ± 0.11 ng DHEA/ng tissue after 72 hours in culture, which was significantly higher when compared with fetal adrenal synthesis of 5.88 ± 0.61 ng DHEA/ng tissue at this time (Figure 6).

Discussion

We undertook this study to establish whether the spiny mouse adrenal gland is capable of synthesizing both DHEA and cortisol and also to examine the localization of key steroidogenic enzymes involved in these processes. We have shown, for the first time, that DHEA is synthesized and secreted by the adrenal gland of the spiny mouse in physiologically significant concentrations in both the prenatal and early neonatal periods. We observed very low levels of circulating corticosterone relative to cortisol in the spiny mouse and can confirm the findings of Lamers et al (23) that plasma cortisol concentrations decrease over

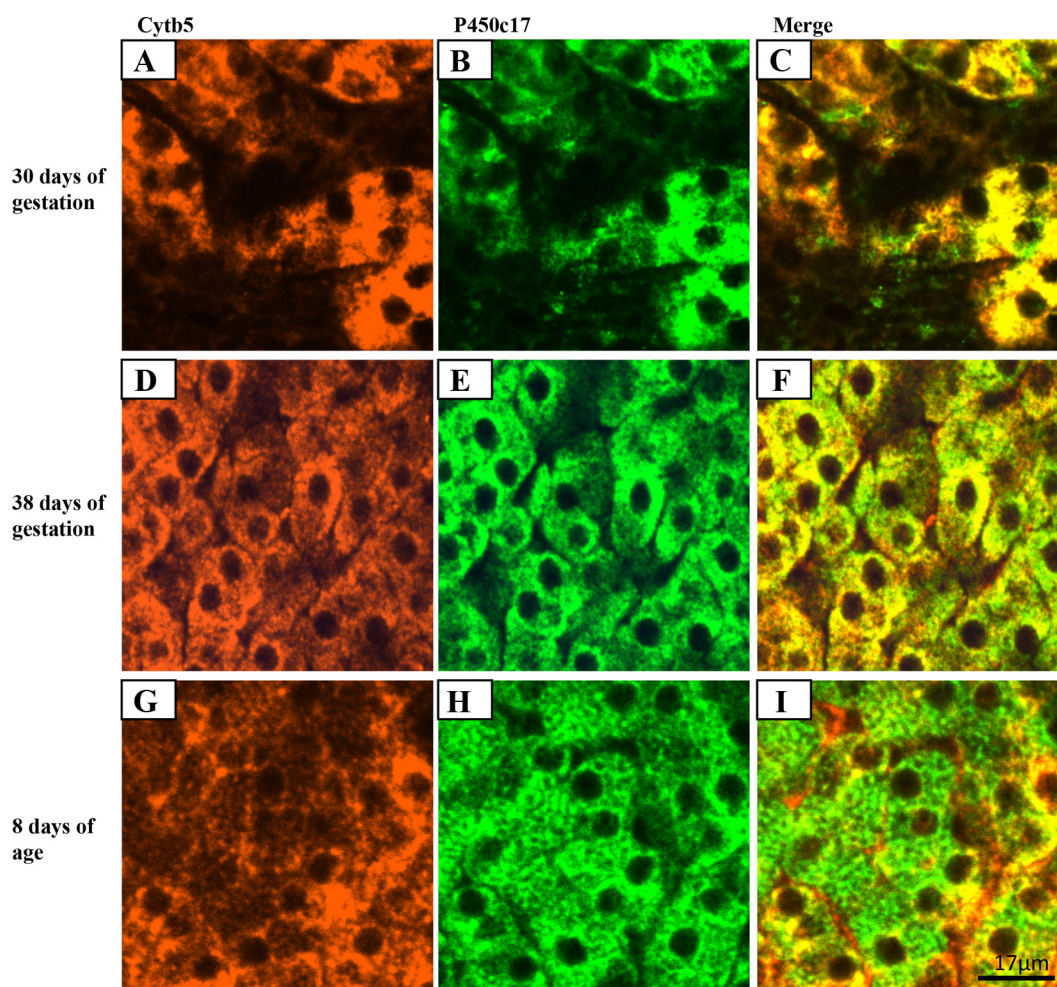


Figure 4. A–I, High-magnification ($\times 400$) photomicrographs showing double-label immunostaining of P450c17- and Cytb5-labeled cells of fetal, 30 (A–C) and 38 (D–F) days of gestation, and postnatal (8 days of age; G–I) male spiny mouse adrenal gland. P450c17 was visualized with Alexa Fluor 488 (green fluorescence), and Cytb5 was visualized with Alexa-Fluor 594 (red fluorescence). Double labeling in the fetal adrenal was distributed mainly in the center of the gland (C), whereas immunopositive cells were primarily localized in the zona reticularis at 38 days of gestation (F). Cells expressing both P450c17 and Cytb5 were localized at the border of the zona reticularis and fasciculata in early postnatal life (I). Cytb5 was found throughout the adrenal gland both in the fetus and adult. Scale bar, 17 μ m.

the last part of the 39-day gestation in this species. We found that the enzyme and accessory protein responsible for the production of DHEA, P450c17 and Cytb5, were coexpressed in the innermost zone of the fetal spiny mouse adrenal cortex and that incubations of whole explant fetal adrenal glands were able to synthesize DHEA from the precursor hormone PREG in a time-dependent manner.

DHEA, cortisol, and aldosterone were detected in fetal plasma from at least 30 days (relative gestation length, 30 days of a 39-day gestation, 0.76) gestation, and both steroids were present throughout prenatal and neonatal life. Although the fetal concentration of DHEA was comparatively low relative to cortisol at 30 days gestation, it increased over the final stages of pregnancy and reached a concentration postnatally similar to that found in human infants between the first and sixth month of life (30). Maternal plasma DHEA concentrations were consistently

higher than fetal concentrations from 30 to 38 days of gestation, suggesting that the maternal and fetal compartments are effectively separate with regard to this hormone.

Explant culture of fetal spiny mouse adrenal gland showed that an increase of exogenous PREG precursor provoked an increase in the secretion of DHEA into the medium. Thus, the P450c17 enzyme identified by immunohistochemistry and by immunoblot from at least 30 days of gestation is likely to be functional, and the DHEA concentrations in the fetal circulation are likely to be of fetal origin. The production of DHEA from PREG, an aspect of adrenal function once considered to be important only in primates (31), shows the capacity of the spiny mouse for extragonadal synthesis and secretion of C_{19} steroids in both fetal and neonatal life, which might subsequently act as precursors for androgen or estrogen synthesis. The regulation of DHEA synthesis and secretion

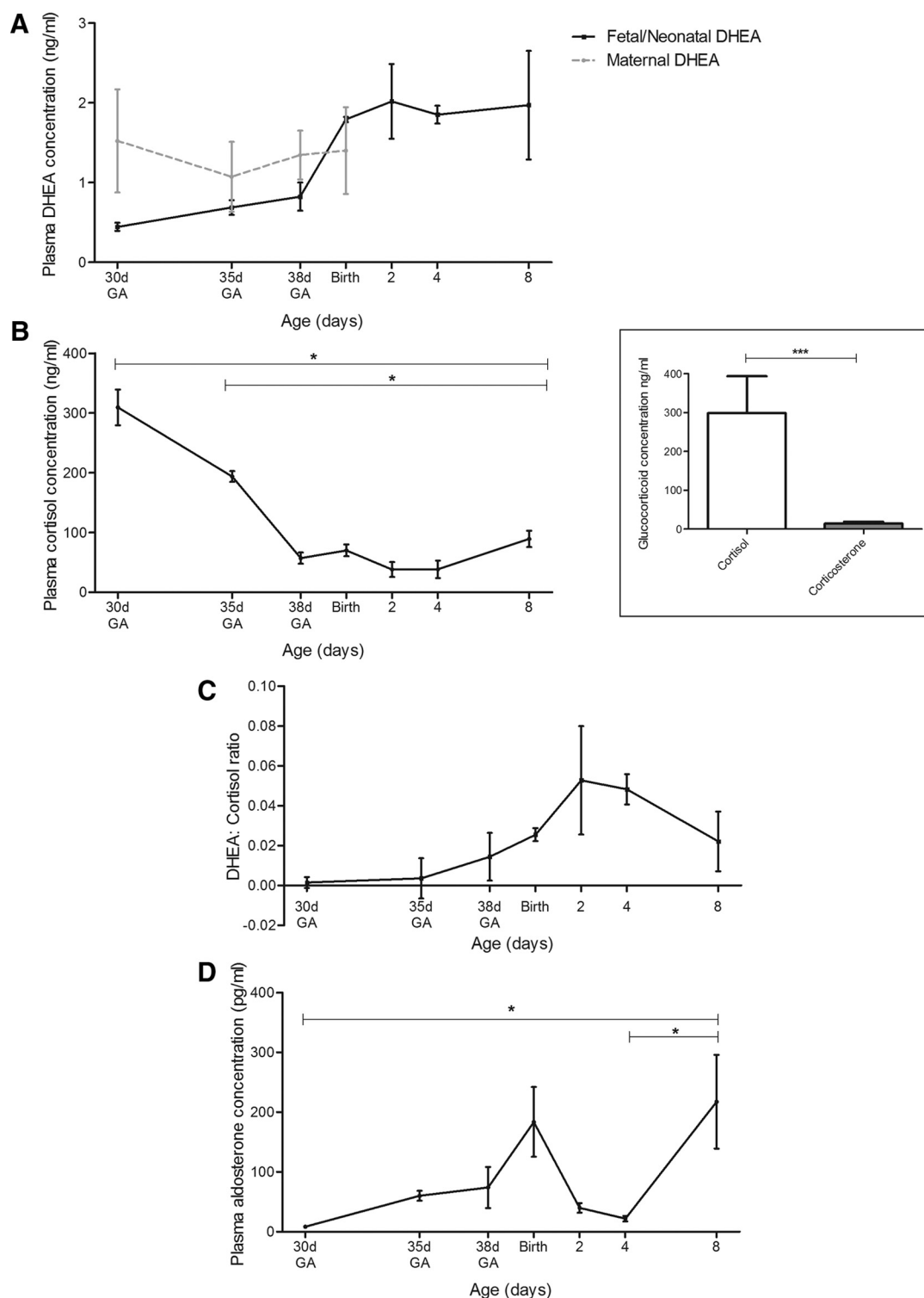


Figure 5. A, Plasma DHEA concentrations in the pregnant spiny mouse dam over the final third of gestation and the day of birth and systemic plasma DHEA concentrations in the fetal and neonatal spiny mouse. The concentrations of DHEA in maternal plasma were highest at 30 days of gestational age (GA). There were no significant differences in maternal DHEA concentrations from this time until the day of birth ($P = .53$). Systemic plasma DHEA levels in the fetus and neonate reached a peak at 8 days of age. B, Plasma cortisol concentrations in the fetal and neonatal spiny mouse. There was a significant decrease in cortisol expression from 30 days of gestation (GA) until birth. There was a significant decrease in cortisol concentrations at 30 and 35 days of gestation, when each was compared with all other prenatal and postnatal ages. The inset in B shows the levels of cortisol relative to corticosterone in spiny mouse plasma at 30 days of gestation. *** $P < .001$; * $P \leq .05$. C, The plasma DHEA to cortisol ratio in the fetal and neonatal spiny mouse. There is an increase in DHEA relative to cortisol concentrations as term approaches, with the peak DHEA to cortisol ratio in the early neonatal period 2 days after birth. D, Plasma aldosterone levels in the fetal and neonatal spiny mouse. Plasma aldosterone levels increased significantly from 30 days of gestation until term and between 4 and 8 days of postnatal age. * $P \leq .05$.

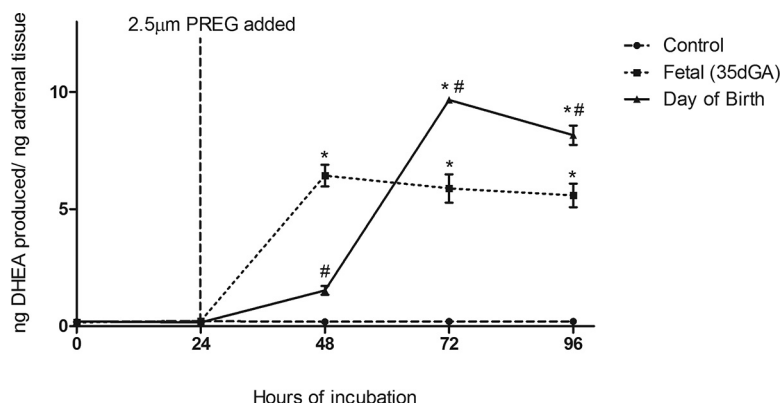


Figure 6. Metabolism of PREG to DHEA by fetal and neonatal adrenal explants as determined by RIA. There were significant main effects of age and time in culture ($P < .0001$, respectively) and a significant interaction effect ($P < .0001$). * $P < .0001$ compared with control (adrenal cultures in the absence of PREG); # $P < .0001$ for fetal adrenal compared with neonatal adrenal. Abbreviation: GA, gestational age.

from the adrenal gland in this species is yet to be determined and, in addition to ACTH, may involve CRH of placental as well as pituitary origin, as shown for the human and other primates (32). Indeed, the decrease of fetal cortisol over the last days of gestation in this species suggests that adrenal DHEA secretion may not be under control of the hypothalamic pituitary adrenal axis alone.

DHEA is produced in large quantities by the fetal zone of the human fetal adrenal gland, due to the abundant expression of the P450c17 enzyme and the accessory hemoprotein Cytb5 (24). The presence of Cytb5 may be indicative of enhanced 17,20-lyase activity in adrenal cells that also express P450c17 (1, 27). DHEA is readily converted into the 3-sulfoconjugate DHEA-S during intra-uterine life, with the circulating concentration of DHEA-S approximately $500\times$ that of DHEA (33). Plasma DHEA-S concentrations reported for the human fetus increase from 34 weeks gestation (1500 ng/mL) until 40 weeks gestation (2500 ng/mL) (34), a pattern of secretion similar to that seen in the pattern of DHEA secretion in the fetal spiny mouse. DHEA concentrations in human neonates have been observed to be around 3.4 ± 1 ng/mL in normal girls and 2.4 ± 0.83 ng/mL in normal boys in the first month of life, falling substantially to around 0.5 ± 0.4 ng/mL in both sexes in the first years of life (35). Thus, the plasma DHEA in the spiny mouse neonate, which was 2 ± 0.46 ng/mL at 2 to 8 days of age, is similar to that of the human infant, suggesting that the adrenal gland is capable of sustained secretion of this C₁₉ steroid after birth in this species.

Although the macroscopic distinction between the cortex and medulla at 30 days of gestation was indistinct, TOH immunohistochemistry indicated that chromaffin-like cells, scattered as small islands of cells throughout the presumptive cortex, were present at this gestational age.

The existence of cortical islands within medullary tissue has also been shown in the rhesus macaque (36) and equine adrenal gland (37) and may not be unusual among mammals. However, it is unclear whether the definitive formation of the adrenal medulla in the spiny mouse comes about as a result of direct preganglionic sympathetic activity, or via the trophic effects of secreted factors of neural origin. The synaptophysin and neurofilament immunohistochemistry indicate that neural elements are present in the medulla at this stage of fetal development. The intermingling of presumptive cortical and medullary cells may be physiologically significant, because DHEA and DHEA-S are known to influence catecholamine synthesis (38, 39), and in vitro administration of DHEA at high concentrations (100 μ M) has been shown to decrease chromaffin progenitor cell number, and when combined with retinoic acid after differentiation induction, increased TOH and dopamine- β -hydroxylase mRNA expression (40). The high expression of Cytb5 observed in the zona glomerulosa and in the medulla of the spiny mouse appears to be a novel finding and requires further investigation. Previous studies of the adult human and primate adrenal gland found that Cytb5 expression was restricted to the zona reticularis (36), the cortical zone responsible for the production of adrenal DHEA (41–43).

In the human fetus, the adrenal glands begin to grow rapidly after the second month of gestation due to hypertrophy of the fetal zone, and the size of the entire gland becomes disproportionately large relative to body weight (44). In the spiny mouse, this appears to occur at approximately 35 days gestation when the rudimentary zonation of the cortex into inner and outer zones occurs; this is also the time when the relative weight of the adrenal gland is the greatest. As the spiny mouse approaches the end of gestation (39 days), the adrenal cortex has differentiated into three distinct zones surrounding a clearly defined medulla.

The common laboratory mouse does not have a functionally distinct zona reticularis because of the lack of P450c17 expression in the adrenal gland (45). A specific feature of the mouse adrenal cortex is the transient X-zone, a putative postpartum remnant of the fetal adrenal zone that regresses at puberty in the male and during the first pregnancy in the female (46). Morphologically, the cytoplasm in the X-zone cells in mice are conspicuously more basophilic than the cytoplasm of cells in the fetal

zone and zona reticularis in humans (47). It is not clear whether this zone is ever present in the prenatal or neonatal spiny mouse. Although there is a zone between the medulla and the cortex after 8 days of postnatal age that is deficient in P450c17, 3 β HSD, and Cytb5 expression (data not shown), cells of the zona reticularis show cytoplasmic eosinophilia and P450c17 immunoreactivity from birth until at least 8 days of age. Indeed, the cortical distribution of P450c17 in the spiny mouse is similar to that described in the guinea pig (48, 49); viz, P450c17 is present in the zonae fasciculata and reticularis, with more intense expression in cells of the outer fasciculata zone and decreasing toward the reticularis cells (48, 49). The high expression of P450c17 in the zona fasciculata compared with the zona reticularis is consistent with the 100-fold greater concentration of cortisol in plasma, compared with the DHEA concentration.

The synthesis and secretion of both cortisol and adrenal-derived androgens also occurs in other small mammals such as the red and Arctic squirrel (50–52) and golden hamsters (53). In adult guinea pigs 11 β -hydroxyandrostenedione is the major C₁₉ steroid secreted into the circulation (54). Although low levels of androstenedione are present in rats during the early neonatal period (2.5 ± 0.98 ng/mL), cortisol concentrations are barely detectable in the rat throughout life (21). De novo synthesis of cortisol requires expression and activity of the 3 β HSD enzyme (55), and in the spiny mouse, we found 3 β HSD expression was greatest in the zona fasciculata, with less intense expression in the outermost zona glomerulosa. Surprisingly, we found expression of 3 β HSD in the innermost cells of the adrenal cortex at 30 but not at 35 days gestation. This suggests that the innermost zone of the adrenal cortex in the fetal spiny mouse may be capable of producing both DHEA and cortisol, and in addition, the coexpression of P450c17, 3 β HSD, and Cytb5 in this zone suggests a capability for androstenedione synthesis and release. From approximately 38 days of gestation, the outermost zone of the cortex expressed 3 β HSD but not P450c17, indicating it may be the primary site of aldosterone synthesis, although this needs to be confirmed by determining the presence of aldosterone synthetase. Consistent with this observation, plasma aldosterone levels in the spiny mouse increased from 38 days until the day of birth. In the human fetal adrenal gland, the outermost definitive zone is relatively inactive early in gestation because of low levels of the mitochondrial cholesterol side-chain cleavage enzyme but is active later in gestation when levels of mineralocorticoids are high, as they are for months after birth (56).

In many species, including pigs, sheep, and humans, fetal plasma cortisol concentration increases markedly before birth (57, 58) and is implicated in the maturation of

fetal organs needed for survival after birth, including the lungs, liver, thyroid, and gut (59). If similar cortisol-dependent maturational events occur in the spiny mouse, it would seem that either essential glucocorticoid-dependent gene expression takes place somewhat before term (around day 35 of the 39-day pregnancy) or another glucocorticoid performs this prepartum role in this species. Although high cortisol concentrations before birth may be necessary for organ maturation (23), they may also compromise the brain where exposure to excess glucocorticoid is known to be neurotoxic (60–62). In this regard, the neuroprotective properties of DHEA may be important, (10, 20, 62–65). The antiglucocorticoid (and hence, neuroprotective) actions of DHEA arise from the ability of this steroid to counteract the toxic effects of high concentrations of corticosterone; eg, it has been shown that even 20nM DHEA is sufficient to increase hippocampal neuronal survival when exposed to 100nM corticosterone (13). Interestingly, in human infant cerebrospinal fluid, there is ~10 times more cortisol than DHEA (66), but the presence of the latter may nevertheless be sufficient to protect against glucocorticoid-induced neurotoxicity during fetal life.

In summary, our study has confirmed that the spiny mouse secretes cortisol as the major circulating glucocorticoid and, importantly, can also synthesize and secrete DHEA in both the fetal and postnatal periods. In this respect, the spiny mouse adrenal gland is similar to the human adrenal gland. Whether DHEA is an essential substrate for placental conversion to estrogens, and has protective and growth-promoting effects in the fetal brain, will require further study.

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